K. Seeliger

seelikat.github.io

Academic positions

May 2025 – present **Martin Luther University Halle-Wittenberg**, *Postdoctoral Research*, Halle.
Developing interpretable oscillatory latent SDE models for neural system identification and forecasting of propofol-induced sedation/consciousness dynamics from large-scale clinical EEG. DFG project leads: Jakob Garbe, Thomas Schmid.

Mar 2020 – Feb 2026 **Max Planck Institute for Human Cognitive and Brain Sciences**, *Postdoctoral Research*, Leipzig.
Deep neural networks and visual neuroscience. End-to-end neural system identification from large-scale fMRI data, learning visual processing models directly from neural responses. Synthesizing preferred stimuli (MEIs) for individual voxels and visual areas inside these models. Group leader: Martin Hebart, since March 2025: Nico Scherf. Guest affiliation since March 2026.

Education

Oct 2015 – Dec 2019 **Ph.D. in Cognitive Computational Neuroscience**, Donders Institute for Brain, Cognition and Behaviour, Nijmegen.
Research on how representations learned by convolutional neural networks relate to human visual processing. Pioneered GAN-based reconstruction of natural images from human brain activity. Recorded what was then the longest single-participant audiovisual fMRI dataset. Used it to develop neural information flow (NIF), the first demonstration of end-to-end system identification across human visual areas directly from large-scale brain data. Supervision: Marcel A. J. van Gerven, Umut Güçlü.

Oct 2012 – Sep 2015 **M.Sc. in Computational Neuroscience**, Technical & Humboldt University, Berlin.
Selective joint degree (offered by TU&HU Berlin) organized by *Bernstein Center for Computational Neuroscience*. My degree included three lab rotations and a seminar project (final grade: 1.4).
Thesis Neural encoding for video stimuli with unsupervised hierarchical representation learning. Supervised by Grégoire Montavon in the group of Klaus-Robert Müller, Machine Learning Group, Technical University of Berlin (grade: 1.0).

Apr 2008 – Apr 2012 **B.Sc. in Computer Science and Media**, Bauhaus University, Weimar.
Computer science degree with research focus through full-semester lab rotations. Courses in experimental and perception psychology (final grade: 1.3).
Thesis Usability of P300-spelling and asynchronous input for text input systems on the *Emotiv consumer EEG*. Supervision: Günther Schatter (grade: 1.3, written: 1.0).

Oct 2007 – Mar 2008 **Media Culture**, Bauhaus University, Weimar.
Transitioned to computer science in first semester.

June 2006 **Abitur**, Albert-Schweitzer-Gymnasium, Bad Düben.
Mathematics / Physics (final grade: 1.4; German Physical Society membership award for best of the year)

Post-graduate education

July 2017 **OCNC**, OIST, Okinawa, OIST Computational Neuroscience Course (fully-funded).

Research visits and projects

Apr 2014 – July 2014 **Center for Information and Neural Networks (CiNet)**, *Research intern*, Osaka, Japan.
Hierarchical video feature learning for an fMRI encoding model for video stimuli (developed into my master thesis).
Supervision: Shinji Nishimoto (ex-gallantlab).

Feb 2014 – Mar 2014 **Max Planck Institute for Mathematics in the Sciences**, *Research intern*, Leipzig.
Extending a network editor for analysing robustness in random boolean networks.
Supervision: Johannes Rauh in the group of Nihat Ay.

Oct 2013 – Jan 2014 **hayneslab, Bernstein Center for Computational Neuroscience**, *Lab rotation*, Berlin.
fMRI pilot experiment for reconstructing spatiotemporal visual perception.
Supervision: Yi Chen and John-Dylan Haynes.

Apr 2013 – Aug 2013 **BBCI Group, Technical University Berlin**, *Seminar research project*, Berlin.
Detecting spatial auditory attention from EEG data in cocktail-party situations.
Supervision: Michael Tangermann in the group of Benjamin Blankertz.

Sep 2010 – Apr 2011 **Cybermedia Center, Osaka University**, *Research intern*, extracurricular, Osaka, Japan.
Large-scale Computational Science and Augmented Reality divisions. Implemented cellular automaton for Abelian sand dune dynamics on a large multi-tile display.
Supervision: Kei Tokita, Macoto Kikuchi and Kiyoshi Kiyokawa.

Apr 2010 – Aug 2010 **Computer Graphics, Bauhaus University**, *Research project*, Weimar.
Setting up a multi-touch display on a sea-container for Bauhaus Summæry.
Supervision: Bernhard Bittdorf.

Mar 2010 – June 2010 **Web Technology & Information Systems, Bauhaus University**, *Research project*, Weimar.
Predicting ⋆ featured articles in Wikipedia – feature engineering for detecting information quality.
Supervision: Maik Anderka in the group of Benno Stein.

Sep 2009 **Digital Games Research Center (DIGAREC)**, *Summer intern*, Potsdam.
Web content revision for the research center and the A MAZE. Berlin gaming festival.

Industry positions

May 2012 – Aug 2012 **Datameer**, *Research & development intern*, San Francisco Bay Area, California.
Implemented a large-scale machine learning model in Hadoop MapReduce, which was subsequently integrated into production with just minor modifications. Supervised by Ulrich Rückert and Hans-Henning Gabriel. Internship supported by GIZ.

July 2007 – Sep 2007 **first site locations**, *Intern, location scouting*, Hamburg.
Photographed and documented locations for advertisement and movie productions. Pre-university media internship.

July 2006 – June 2007 **Rockstar Games**, *German localisation tester*, Lincoln, UK.
General quality assurance, proofreading and improvement of translations, and game design evaluation. Worked on Canis Canem Edit (Bully), The Warriors, GTA: Vice City Stories, and Manhunt 2.

Other positions

Mar 2017 – Sep 2018 **Donders Center for Cognition (DCC)**, *Software management & support*, Nijmegen.
DCC Cluster admin work and support.

Mar 2013 – Aug 2015 **German Aerospace Center (DLR), Institute of Planetary Research**, *Student assistant*, Berlin.
Machine learning-based visual smoke detection for *FireWatch*; developed a Python control wrapper for the pco.edge sCMOS camera; ran particle simulations on the HLRN HPC system for *Rosetta* comet visualizations; developed a metadata database and Python GUI for *Rosetta* OSIRIS, VIRTIS, and NAVCAM data. Supervised by Stubbe Hviid and Ekkehard Kührt in the *Asteroids and Comets* department.

June 2010 – Aug 2012 **Web Technology & Information Systems, Bauhaus University**, *Student assistant*, Weimar.
Large-scale information retrieval and feature engineering in the Wikipedia revision history with Hadoop MapReduce; supervised by Maik Anderka.

Aug 2009 – Apr 2010 **Building Physics, Bauhaus University**, *Student assistant*, Weimar.
Literature research for Building Physics eLearning games; supervised by Thomas Bröker and Heinrich Söbke.

Technical skills

Programming: Python (primary; scientific computing and ML: NumPy, SciPy, PyTorch, Chainer, scikit-learn), C++, MATLAB, Java.

Additional: R, Ada, Scheme, JavaScript (project/coursework experience).

ML / AI: Deep learning (CNNs, autoencoders, generative models), representation learning, large-scale neural system identification, time-series modelling, latent dynamical models (including SDEs).

Neuroscience: Neural encoding and decoding models, fMRI/MEG/EEG analysis, visual perception reconstruction, representational similarity analysis, receptive field modelling, spectral analysis.

Tools: Linux/Unix command line, Git, HPC and cluster administration, FieldTrip, Hadoop MapReduce.

Markup: LaTeX, HTML, XML.

Teaching activities

2026 *Representation Learning & Autoencoders* as part of the Deep Learning Lab lecture (Martin Luther-University Halle-Wittenberg).

2025 *Introduction to neuroAI* as part of the Artificial Intelligence lecture (MPI-CBS, Leipzig University).

2024 *Neural encoding and RSA* workshop at *IICCSSS* summer school (Osnabrück). Prepared and taught RSA and encoding model workshop.

2022 *Deep Learning and Vision* session at *IMPRS NeuroCom* summer school (Leipzig). Prepared and taught RSA practical.

2016 – 2018 Tutor Artificial Neural Networks (AI B.Sc.). Preparing, assisting and grading the assignments and the final exam. One of two tutors. My final tutoring year 2018 received an evaluation which was very high (8.1).

2017 – 2018 Tutor Python for Artificial Intelligence (AI B.Sc.). Assisting and grading assignments. One of four tutors.

2017 Tutor Academic and Professional Skills (AI B.Sc.). Grading final essays. One of four tutors.

2016 *Computational Modeling and Cognitive Development* at *Amsterdam Brain and Cognition Summer School*. Prepared and assisted one-week practical.

Supervision

Master theses (individual supervision)

Alexander Hergett — Image representation analysis using SPoSE in the higher ventral visual cortex, 2023, Leipzig University.

Roman Leipe — End-to-end learning in artificial neural networks for accurate prediction of visually evoked functional MRI Data, 2023, Leipzig University.

Johannes Roth — Generating most exciting images for neural populations via image synthesis from NIF Models, 2021, Leipzig University.

Leonieke van den Bulk — Reconstructing naturalistic movies from fMRI brain responses: Comparing motion-energy features with convolutional neural network representations, 2019, Radboud University.

Elizaveta Genke — Functional specialization in the ventral visual stream examined with convolutional neural network-derived visual representations, 2019, Erasmus Mundus MAIA.

Marjolein Troost — Generalization of an upper bound on the number of nodes needed to achieve linear separability, 2018, Radboud University.

Hugo Thabo Dictus — Encoding deep predictive network error responses as a representation of fMRI responses in visual areas, 2018, VU & University of Amsterdam.

Secondary reader

Master theses — Sebastian Tiesmeyer (2019, Radboud University), Mart van Rijthoven (2018, Radboud University)

Bachelor theses — Lino Vliex (2016, Radboud University)

Internships

Florian Mahner — Generative adversarial networks for video generation, 2018, Osnabrück & Radboud University.

Reviewing

primary — CCN meeting, Human Brain Mapping, Nature Communications, npj Digital Medicine, Nature Machine Intelligence, Neurons, Behavior, Data analysis & Theory, NeurIPS SVRHM, ICLR Re-Align, PLoS Computational Biology, PLoS Biology

co-reviews — Nature Human Behaviour, NeurIPS, Current Biology, Journal of Neuroscience

Active collaborations

- Antonio Lozano & Paolo Papale of Roelfsema Group, Netherlands Institute for Neuroscience, Amsterdam, The Netherlands and University Miguel Hernández of Elche, Spain. *Topic*: Computational models for visual prostheses.
- Julie Boyle, Lune Bellec & Marie St-Laurent, Courtois NeuroMod project, University of Montréal, Canada. *Topic*: Large-scale data, end-to-end learning.
- Affiliated with Neural Coding Lab of Umut Güçlü, Donders Institute, Nijmegen, The Netherlands.

Organizational activities

CoCoNUT — Co-organizer of the Cognitive Computational Neuroscience Unification Trial talk series at MPI-CBS (2020–).

DOSE-I Challenge — Organizer of the ICASSP 2027 Signal Processing Grand Challenge, *Forecasting Responsiveness under Propofol*. Designed / implemented challenge setup, benchmark design, dataset splits, starter kit, baselines/evaluation; participant support (2026–2027).

Octagon — Organizing team of a student-run internet café at Albert-Schweitzer-Gymnasium. Organized concerts, movie nights, pen-and-paper RPG sessions, etc.; regular staffing and technical support (1998–2006).

Publications

2026 Garbe, J., Q. V. Nguyen, J. W. Kantelhardt, F. D., K. Erffmeier, **K. Seeliger**, and T. Schmid (2026). "Towards Predicting Sedation Depth in Endoscopy with Large Clinically Annotated EEG Data of Continuous Propofol Sedation". In: *Artificial Intelligence in Medicine: 24th International Conference, AIME 2026, Ottawa, ON, Canada, July 7–10, 2026, Proceedings*. Lecture Notes in Computer Science. Accepted / in press. Springer.

Gözükara, Dora, Nasir Ahmad, **K. Seeliger**, Djamari Oetringer, and Linda Geerligs (2026). "Neural network-based encoding in free-viewing fMRI with gaze-aware models". In: *Neurons, Behavior, Data analysis, and Theory* 9.1. DOI: 10.51628/001c.158956. arXiv: 2603.11663 [q-bio.NC].

St-Laurent, Marie, Basile Pinsard, Oliver Contier, Elizabeth DuPre, **K. Seeliger**, Valentina Borghesani, Julie A. Boyle, Lune Bellec, and Martin N. Hebart (2026). "CNeuroMod-THINGS, a densely-sampled fMRI dataset for visual neuroscience". In: *Scientific Data* 13.1, p. 141. DOI: 10.1038/s41597-026-06591-y.

Lozano*, Antonio, **K. Seeliger***, Aleksandr Kovalev, Umut Güçlü, and Yağmur Güçlütürk (2026). "Neuroprosthetics From and Beyond Restoration: Sensory Recovery, Adaptive Interfaces and the Path to Augmentation". In: *Proceedings of the Augmented Humans International Conference 2026*. Association for Computing Machinery, pp. 741–743. DOI: 10.1145/3795011.3797159.

2025 Le, Lynn, Thirza Dado, **K. Seeliger**, Paolo Papale, Antonio Lozano, Pieter Roelfsema, Yağmur Güçlütürk, Marcel van Gerven, and Umut Güçlü (2025). "Inverse receptive field attention for naturalistic image reconstruction from the brain". In: *arXiv preprint arXiv:2501.03051*. DOI: 10.48550/arXiv.2501.03051. arXiv: 2501.03051 [q-bio.NC].

Le, Lynn, Nils Kimman, Thirza Dado, **K. Seeliger**, Paolo Papale, Antonio Lozano, Pieter Roelfsema, Marcel van Gerven, Yağmur Güçlütürk, and Umut Güçlü (2025). "Neural encoding with affine feature response transforms". In: *arXiv preprint arXiv:2501.03741*. DOI: 10.48550/arXiv.2501.03741. arXiv: 2501.03741 [q-bio.NC].

2024 Le, Lynn, Paolo Papale, **K. Seeliger**, Antonio Lozano, Thirza Dado, Feng Wang, Marcel van Gerven Pieter R. Roelfsema, Y. Güçlütürk, and U. Güçlü (2024). "MonkeySee: Space-time-resolved reconstructions of natural images from macaque multi-unit activity". In: *Advances in Neural Information Processing Systems (NeurIPS) 2024*.

Seeliger, K. and M. N. Hebart (2024). "What comparing deep neural networks can teach us about human vision". In: *Nature Machine Intelligence* 6.2, pp. 122–123. DOI: 10.1038/s42256-024-00789-8.

2023 Doerig*, A., R. P. Sommers*, **K. Seeliger**, B. Richards, J. Ismael, G. W. Lindsay, K. P. Körding, T. Konkle, M. A. J. van Gerven, N. Kriegeskorte, and T. C. Kietzmann (2023). "The neuroconnectionist research programme". In: *Nature Reviews Neuroscience* 24.7, pp. 431–450. DOI: 10.1038/s41583-023-00705-w.

Golan, T., J. M. Taylor, H. Schütt, B. Peters, R. P. Sommers, **K. Seeliger**, A. Doerig, P. Linton, T. Konkle, M. A. J. van Gerven, K. P. Körding, B. Richards, T. C. Kietzmann, G. W. Lindsay, and N. Kriegeskorte (2023). "Deep neural networks are not a single hypothesis but a language for expressing computational hypotheses". In: *Behavioral and Brain Sciences* 46, e392. DOI: 10.1017/S0140525X23001553.

2022 Le, L., L. Ambrogioni, K. **Seeliger**, Y. Güçlütürk, M. A. J. van Gerven, and U. Güçlü (2022). "Brain2Pix: Fully convolutional naturalistic video reconstruction from brain activity". In: *Frontiers in Neuroscience*.

Singer, J.D., K. **Seeliger**, T. C. Kietzmann, and M. N. Hebart (2022). "From photos to sketches-how humans and deep neural networks process objects across different levels of visual abstraction". In: *Journal of Vision* 22.2, pp. 4–4.

2021 **Seeliger**, K., L. Ambrogioni, Y. Güçlütürk, L. M. van den Bulk, U. Güçlü, and M. A. J. van Gerven (2021). "End-to-end neural system identification with neural information flow". In: *PLOS Computational Biology* 17.2.

2019 Gerven, M. A. J. van, K. **Seeliger**, U. Güçlü, and Y. Güclütürk (2019). "Current advances in neural decoding". In: *Explainable AI: Interpreting, Explaining and Visualizing Deep Learning*. Springer, pp. 379–394.

Seeliger*, K., R. P. Sommers*, U. Güçlü, S. E. Bosch, and M. A. J. van Gerven (2019). "A large single-participant fMRI dataset for probing brain responses to naturalistic stimuli in space and time". In: *bioRxiv preprint*.

2018 Rijthoven, M. van, Z. Swiderska-Chadaj, K. **Seeliger**, J. van der Laak, and F. Ciompi (2018). "You only look on lymphocytes once". In: *Medical Imaging with Deep Learning (MIDL) 2018*.

Seeliger, K., M. Fritsche, U. Güçlü, S. Schoenmakers, J.-M. Schoffelen, S. E. Bosch, and M. A. J. van Gerven (2018). "Convolutional neural network-based encoding and decoding of visual object recognition in space and time". In: *NeuroImage* 180, pp. 253–266.

Seeliger, K., U. Güçlü, L. Ambrogioni, Y. Güçlütürk, and M. A. J. van Gerven (2018). "Generative adversarial networks for reconstructing natural images from brain activity". In: *NeuroImage* 181, pp. 775–785.

Troost, M., K. **Seeliger**, and M. A. J. van Gerven (2018). "Generalization of an upper bound on the number of nodes needed to achieve linear separability". In: *Benelux Conference on Artificial Intelligence (BNAIC) 2018*.

2017 Güçlütürk, Y., U. Güçlü, K. **Seeliger**, S. Bosch, E. van Lier, and M. A. J. van Gerven (2017). "Reconstructing perceived faces from brain activations with deep adversarial neural decoding". In: *Advances in Neural Information Processing Systems (NeurIPS) 2017*, pp. 4249–4260.

2016 Bosch, S., K. **Seeliger**, and M. A. J. van Gerven (2016). "Modeling Cognitive Processes with Neural Reinforcement Learning". In: *bioRxiv preprint*.

Datasets and other resources

2026 Garbe, Jakob, Quang Vu Nguyen, Jan W. Kantelhardt, Florian Dünninghaus, Karla Erffmeier, Katja Seeliger, and Thomas Schmid (2026). *DOSE-I: A Multimodal Biosignal Dataset of Procedural Sedation for Endoscopy*. Zenodo. Dataset. DOI: 10.5281/zenodo.18483292.

Seeliger, **K.** (2026). *Neuro-Visual Reconstruction Dataset Index*. Version v1.1. Curated dataset index. DOI: 10.5281/zenodo.18876185. URL: https://github.com/seelikat/neuro-visual-reconstruction-dataset-index.

2025 St-Laurent, Marie, Basile Pinsard, Olivier Contier, Elizabeth DuPre, Katja Seeliger, Valentina Borghesani, Julie A. Boyle, Lune Bellec, and Martin N. Hebart (2025). *cneuromod-things*. Zenodo. DOI: 10.5281/zenodo.17881592.

2024 *Naturalistic walking-tour video stimuli and macaque neural recordings* (2024). Roelfsema Lab, Netherlands Institute for Neuroscience, Amsterdam. Not yet published.

2018 Seeliger, Katja, Rowan P. Sommers, Umut Güçlü, Sander E. Bosch, and Marcel A. J. van Gerven (2018). *The Doctor Who dataset: A large single-participant fMRI dataset for probing brain responses to naturalistic stimuli in space and time*. Radboud Data Repository. Dataset. DOI: 10.34973/j05g-fr58.

Talks

Invited talks

2026 *System identification from raw EEG under clinical propofol sedation with an interpretable latent SDE model. Consciousness Journal Club.* MPI-CBS & MPI-MIS, Leipzig, Germany.
Host: Samson Odan.

Deep Neural Networks in Sensory Neuroscience: Representation, Similarity, and the Irreducibility of Training. Workshop on *Philosophical Issues in Neural Computation*. Ruhr University Bochum, Germany.
Hosts: Nikola Kompa, Tobias Schlicht & Johannes Brinz.

2025 *Extracting neural information processing systems from large neuroimaging data.* BCCN Alumni symposium. Bernstein Center for Computational Neuroscience, Berlin, Germany.

2024 *Leveraging massive fMRI data sets and deep learning to synthesize images preferred by higher visual system areas.* The Brain Mosaic: Cellular Heterogeneity in the CNS, 3rd edition. VIB.AI, Leuven, Belgium.

In-silico modelling of visual information processing. VIB.AI, Leuven, Belgium.
Host: Stein Aerts.

Extracting neural information processing systems from large neuroimaging data. Workshop on Hybrid AI Networks. Lancaster University, Leipzig, Germany.

2023 *Leveraging massive fMRI data sets and deep learning to synthesize images preferred by higher visual system areas.* Guest talk at lab retreat. FU Berlin, Neural Dynamics of Visual Cognition, Germany.
Host: Johannes Singer.

Convolutional neural networks and human visual information processing. 1st Leipzig Symposium on Intelligent Systems. Leipzig University, Germany.

2022 *Leveraging large data sets and deep learning to synthesize images preferred by brain areas.* Symposium on Deep Learning and Vision Neuroscience. Iberian Conference on Perception (CIP), Barcelona, Spain.

2021 *Convolutional neural networks and visual information processing.* Search Symposium Computational Neuroscience. Osnabrück University, virtual.

2019 *Uncovering visual system representations with convolutional neural networks.* Max Planck Institute for Human Cognitive and Brain Sciences, Vision and Computational Cognition Group, Leipzig, Germany.
Host: Martin Hebart.

End-to-end learning of neural information processing systems from brain data. Deep Learning in Computational Neuroscience Workshop. Bernstein Conference, Berlin, Germany.

Neural network representations and visual processing in brains. Symposium on Big Data in Vision Science. European Conference on Visual Perception (ECVP), Leuven, Belgium.

2017 *Neural network representations and visual processing in brains.* Schloss Dagstuhl Seminar on Human-Like Neural-Symbolic Computing. Wadern, Germany.

Panel discussions

2024 *Roundtable: The Future of Neurotechnology.* Dutch Brain Consortium (DBI^2) and INTENSE joint network and outreach event. Het Trippenhuis, Amsterdam, The Netherlands.

Contributed talks

2021 *A large single-participant fMRI dataset for probing brain responses to naturalistic stimuli in space and time.* Symposium on Naturalistic Stimuli in Cognitive Neuroscience. Tagung experimentell arbeitender Psychologen (TeaP), virtual.

Invited seminars

2026 *Neural Digital Twins.* University of Göttingen, Germany.
Host: Fabian Sinz.

2023 *Leveraging massive fMRI data sets and deep learning to synthesize images preferred by higher visual system areas.* Netherlands Institute for Neuroscience, Roelfsema Group, Amsterdam, The Netherlands.
Host: Antonio Lozano.

2022 *ConvNets, end-to-end learning and visual information processing.* Research seminar. King Lab at META, virtual.
Host: Jean-Rémi King.

Leveraging massive fMRI data sets and deep learning to synthesize images preferred by higher visual system areas. Lau Lab, virtual.
Host: Hakwan Lau.

2021 *Research talk.* Project seminar. University of Montreal, Courtois NeuroMod project, virtual.
Hosts: Julie A. Boyle & Pierre Bellec.

Conference posters

2026 K. Seeliger, J. Kantelhardt, J. Garbe, N. Scherf, T. Schmid: System identification from raw EEG during clinical propofol sedation with an interpretable latent SDE model. Conference on Cognitive Computational Neuroscience (CCN) 2026, New York.

J. Garbe, F. Dünninghaus, J. W. Kantelhardt, J. Rosendahl, K. Seeliger, T. Schmid: *Eine neue Datengrundlage für die translationale Erforschung der Endoskopie-Sedierung: der DOSE-I Biosignaldatensatz mit klinischer Echtzeit-Annotation.* Zeitschrift für Gastroenterologie 64(8), KV–354, 2026, conference abstract/poster. DOI: 10.1055/s-0046-1826818.

2023 K. Seeliger, R. Leipe, J. Roth, M. N. Hebart: Uncovering high-level visual cortex preferences by training convolutional neural networks on large neuroimaging data. Neuro-AI-Talks (NEAT), Osnabrück, Germany, poster.

2023 K. Seeliger, R. Leipe, J. Roth, M. N. Hebart: Investigating High-Level Visual Cortex Preferences through Neural Network Training on Large Neuroimaging Data. Salzburg Mind Brain Meeting (SAMBA) 2023, Salzburg, Austria.

K. Seeliger, R. Leipe, J. Roth, M. N. Hebart: Uncovering high-level visual cortex preferences by training convolutional neural networks on large neuroimaging data. Vision Sciences Society (VSS) 2023, St. Pete Beach, Florida.

M. St-Laurent, B. Pinsard, O. Contier, K. Seeliger, V. Borghesani, J. Boyle, P. Bellec, M. N. Hebart: `cneuromod-things`: A large-scale fMRI dataset for task- and data-driven assessment of object representation and visual memory recognition in the human brain. Vision Sciences Society (VSS) 2023, St. Pete Beach, Florida.

O. Contier, S. Fujimori, K. Seeliger, N. A. Ratan Murty, M. N. Hebart: Revealing interpretable object dimensions from a high-throughput model of the fusiform face area. 23rd Annual Meeting of the Vision Science Society (VSS), St. Pete Beach, FL, USA, poster.

2022 W. Kłos, P. Coronica, K. Seeliger, M. N. Hebart: Training BigGAN on an ecologically motivated image dataset. Conference on Cognitive Computational Neuroscience (CCN), San Francisco, California.

F. P. Mahner, K. Seeliger, U. Güçlü, M. N. Hebart: Learning Cortical Magnification with Brain-Optimized Convolutional Neural Networks. Conference on Cognitive Computational Neuroscience (CCN), San Francisco, California.

M. St-Laurent, K. Seeliger, M. Hebart: The role of gaze position in training visual brain encoders on free-viewing data. Vision Sciences Society (VSS), 2022, St. Pete Beach, Florida.

2021 K. Seeliger, J. Roth, M. N. Hebart: Synthesizing preferred stimuli for individual voxels in the human visual system. Vision Sciences Society (V-VSS), Virtual, poster.

J. Roth, K. Seeliger, T. Schmid, M. N. Hebart: Synthesizing Preferred Stimuli for Individual Voxels in the Human Visual System. Computational and Systems Neuroscience (COSYNE), 2021 (virtual).

2019 J. Singer, K. Seeliger, M. N. Hebart: The representation of object drawings and sketches in deep convolutional neural networks. NeurIPS workshop on Shared Visual Representations in Human & Machine Intelligence (SVRHM), 2020 (virtual).

K. Seeliger, L. Ambrogioni, U. Güçlü, M. A. J. van Gerven: Neural Information Flow: Learning neural information processing systems from brain activity. Conference on Cognitive Computational Neuroscience (CCN), Berlin.

2017 K. Seeliger, U. Güçlü, L. Ambrogioni, Y. Güçlütürk, and M. A. J. van Gerven: *Reconstructing natural images from brain data using pretrained deep generative adversarial networks.* Accepted poster presentation; not presented due to travel-funding constraints. Inaugural Conference on Cognitive Computational Neuroscience (CCN), Columbia University, New York City, USA.

2016 K. Seeliger, M. Fritsche, U. Güçlü, S. Schoenmakers, J.-M. Schoffelen, S. E. Bosch and M.Ã.J. van Gerven: A forward pass through the visual system: ConvNets encode MEG source activity. NeurIPS workshop MLINI, Barcelona.

K. Seeliger, G. Montavon, K.-R. Müller, M. A. J. van Gerven, S. Nishimoto: Hierarchical K-means encodes human visual cortex activity on video stimuli. NeurIPS workshop MLINI, Barcelona.

S. E. Bosch, K. Seeliger, M. A. J. van Gerven: Modeling human probabilistic categorization with neural reinforcement learning. NeurIPS workshop Brains+Bits, Barcelona.

K. Seeliger, M. Fritsche, U. Güçlü, S. Bosch, S. Schoenmakers and M. A. J. van Gerven: Convolutional neural networks code for spatiotemporal MEG source activity across the visual system. ICT.OPEN2016, Amersfoort.

2014 M. Tangermann, K. Seeliger, A. Nolte, J. Schumacher, P. Zhutovsky, B. Blankertz: Detecting spatial auditory attention in cocktail-party situations. 30th International Congress of Clinical Neurophysiology (ICCN) of the IFCN, Berlin.

Miscellaneous

2024 7T SIEMENS MAGNETOM TERRA self-scanning-certified (Max Planck Institute for Human Cognitive and Brain Sciences, certified by DOMENICA KLANK and ROBERT TRAMPEL).

2017 3T SIEMENS MAGNETOM PRISMA scanning-certified (Donders Centre for Cognitive Neuroimaging, certified by PAUL GAALMAN).

Languages English (fluent), German (native), Japanese (CEFR B1), Dutch/Flemish (CEFR A2), French (CEFR A1).

References

Prof. Dr. Klaus-Robert Müller
Machine Learning Group
Technical University Berlin
klaus-robert.mueller@tu-berlin.de

Prof. Dr. Shinji Nishimoto
PI Perceptual and Cognitive Neuroscience Lab
Center for Information and Neural Networks
(CiNet, Osaka University)
nishimoto@nict.go.jp

Prof. Dr. Martin N. Hebart
Professor for Computational Cognitive
Neuroscience and Quantitative Psychiatry
Justus-Liebig University Giessen
office-hebart@med.uni-giessen.de

Prof. Dr. Umut Güçlü
PI Neural Coding Lab
Radboud University, Donders Institute
for Brain, Cognition and Behaviour
u.guclu@donders.ru.nl

Prof. Dr. Marcel A. J. van Gerven
PI Artificial Cognitive Systems lab
Radboud University, Donders Institute
for Brain, Cognition and Behaviour
marcel.vangerven@donders.ru.nl

Dr. Nico Scherf
Group Leader, Neural Data Science and
Statistical Computing
Max Planck Institute for Human Cognitive
and Brain Sciences (MPI-CBS) Leipzig
nscherf@cbs.mpg.de